

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-220***

“Price 25 kV”

December 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Urban Grid Solar Projects, LLC, the Interconnection Customer (IC), has proposed a 26.2 MW (15.4 MWC) solar generating facility to be located in Queen Anne's County, Maryland. PJM studied the AC1-220 project into the Delmarva Power and Light Company (DPL) system at a tap of the Price-IB Corners 69 kV circuit and evaluated it for compliance with reliability criteria for summer peak conditions in 2020. The planned in-service date, as requested by the IC, is September 30, 2018. This date may not be attainable due to additional required PJM studies and the Transmission Owner construction schedule.

Point of Interconnection

The Interconnection Customer requested a 69 kV transmission level interconnection. As a result, AC1-220 will require a Point of Interconnection (POI) off DPL's 69 kV Price – IB Corners circuit 6710 in Queen Anne's County, Maryland, approximately 1.5 miles from Price Substation. The POI will require the creation of a new 69 kV ring bus substation. The new substation will include two terminals for Circuit 6710 and one terminal for AC1-220, along with required relaying and protection (see Attachment 1).

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Transmission Owner Scope of Work

Substation Interconnection Estimate

Scope: Build a new 69 kV substation with a 3 position ring bus. Two of the positions on the ring bus will be transmission line terminals for the tie-in of Line 6710 to the substation. The other position will be a terminal configured for the interconnection of the AC1-220 generator. This will include adding 3 new 69 kV circuit breakers, disconnect switches, CVTs, line relays, breaker relays, and associated bus equipment and support structures.

Estimate: \$4,750,000

Construction Time: 24-36 months, plus any outage delays

Major Equipment Included in Estimate:

• Control Enclosure, 20' x 15'	Qty. 1
• Power Circuit Breaker, 69 kV, 2000A, 40kA, 3 cycle	Qty. 3
• Disconnect Switch, 69 kV, 2000A, Manual Wormgear, Arcing Horns	Qty. 9
• CT/VT Combination Units, 69 kV	Qty. 3
• CVT, 69 kV	Qty. 6
• Disconnect Switch Stand, High, 69 kV, Steel	Qty. 5
• Disconnect Switch Stand, Low, 69 kV, Steel	Qty. 4
• CT/VT Stand, Single Phase, Low, 69 kV, Steel	Qty. 3
• CVT Stand, Single Phase, Low, 69 kV, Steel	Qty. 6
• SSVT, 69 kV/240-120 V	Qty. 1
• Relay Panel, Transmission Line, FL/BU (20")	Qty. 3
• Control Panel, 69 kV Circuit Breaker (10")	Qty. 3
• Take-off structure, 69 kV	Qty. 2
• Bus Support Structure, 3 phase, 69 kV, Steel	Qty. 8

Estimate Assumptions:

- Land purchase for the substation is not included.
- A 3.0 acre, relatively square lot is available for use.
- Site clearing and grading performed by Developer.
- Storm water management is required, as a disturbance of greater than 5000 square feet requires mitigation
- Lightning protection (lightning masts) are not required.
- Permitting will be performed by DPL.

Transmission Engineering Interconnection Estimate

Scope: Reconfigure Line 6710 to accommodate the new terminal positions at the new AC1-220 interconnection substation.

Estimate: \$350,000

Construction Time: 24 months

Required Relaying and Communications

New protection relays are required for the new generator terminal. An SEL-487 will be required for primary protection and an SEL-387 will be required for back-up protection. A 20" relay panel is required.

New protection relays are required for the new line terminals. An SEL-487 will be required for primary protection and an SEL-387 will be required for back-up protection. A 20" relay panel is required (2 total).

A SEL-451 relay on a 20" breaker control panel will be required for the control and operation of the new 69 kV circuit breakers (3 total).

An Ethernet switch will be required to interface the new protection relays with the existing Orion data concentrator. New equipment will need to be installed and tested.

The project will require re-wiring and adjustment of existing relay schemes to accommodate the new 69 kV substation.

Relay replacements at the remote ends of Circuit 6710 at Church Substation and Price Substation. Relays will be installed for primary and back-up protection. The relay replacements at both ends will cost approximately **\$400,000** and will require **24 months**, plus any outage delays, to complete.

Metering

Three phase 69 kV revenue metering points will need to be established. DPL will purchase and install all metering instrument transformers as well as construct a metering structure. The secondary wiring connections at the instrument transformers will be completed by DPL's metering technicians. The metering control cable and meter cabinets will be supplied and installed by DPL. DPL will install conduit for the control cable between the instrument transformers and the metering enclosure. The location of the metering enclosure will be determined in the construction phase. DPL will provide both the Primary and the Backup meters. DPL's meter technicians will program and install the Primary & Backup solid state multi-function meters for each new metering position. Each meter will be equipped with load profile, telemetry, and DNP outputs. The IC will be provided with one meter DNP output for each meter. DPL will own the metering equipment for the interconnection point, unless the IC asserts its right to install, own, and operate the metering system.

The Interconnection Customer will be required to make provisions for a voice quality phone line within approximately 3 feet of each Company metering position to facilitate remote interrogation and data collection.

It is the IC's responsibility to send the data that PJM and DPL requires directly to PJM. The IC will grant permission for PJM to send DPL the following telemetry that the IC

sends to PJM: real time MW, MVAR, volts, amperes, generator status, and interval MWH and MVARH. The estimate for DPL to design, purchase, and install metering as specified in the aforementioned scope for metering is included in the Substation Interconnection Estimate.

Interconnection Customer Scope of Work

The Interconnection Customer is responsible for all design and construction related to activities on their side of the Point of Interconnection. Site preparation, including grading and an access road, as necessary, is assumed to be by the IC. Route selection, line design, and right-of-way acquisition of the direct connect facilities is not included in this report, and is the responsibility of the IC. Protective relaying and metering design and installation must comply with DPL's applicable standards. The IC is also required to provide revenue metering and real-time telemetering data to PJM in conformance with the requirements contained in PJM Manuals M-01 and M-14 and the PJM Tariff.

DPL Interconnection Customer Scope of Direct Connection Work Requirements:

- DPL requires that an IC circuit breaker is located within 500 feet of the DPL substation to facilitate the relay protection scheme between DPL and the IC at the Point of Interconnection (POI).

Additional Operating Requirements

1. The Company (DPL, ACE, Pepco) will require the capability to remotely disconnect the generator from the grid by communication from its System Operations facility. This will be accomplished with a line recloser.
2. Company reserves the right to charge the Interconnection Customer operation and maintenance expenses to maintain the Interconnection Customer attachment facilities, including metering and telecommunications facilities, owned by Company.

Summer Peak Analysis - 2020

Transmission Network Impacts

Potential transmission network impacts are as follows:

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

1. (DP&L - DP&L) The MIDLTNTP-MT PLSNT 138 kV line (from bus 232106 to bus 232104 ckt 1) loads from 100.85% to 103.49% (AC power flow) of its emergency rating (348 MVA) for the tower line contingency outage of 'DBL_4NC'. This project contributes approximately 10.12 MW to the thermal violation.

CONTINGENCY 'DBL_4NC'/* RED LION-CEDAR CREEK 230;RED LION-CARTANZA 230
OPEN LINE FROM BUS 231004 TO BUS 232002 CKT 1
OPEN LINE FROM BUS 231004 TO BUS 232003 CKT 1
END

Please refer to Appendix 1 for a table containing the generators having contribution to this flowgate.

2. (DP&L - DP&L) The TOWNSEND-MIDLTNTP 138 kV line (from bus 232107 to bus 232106 ckt 1) loads from 111.18% to 113.81% (AC power flow) of its emergency rating (348 MVA) for the tower line contingency outage of 'DBL_4NC'. This project contributes approximately 10.12 MW to the thermal violation.

CONTINGENCY 'DBL_4NC'/* RED LION-CEDAR CREEK 230;RED LION-CARTANZA 230
OPEN LINE FROM BUS 231004 TO BUS 232002 CKT 1
OPEN LINE FROM BUS 231004 TO BUS 232003 CKT 1
END

Please refer to Appendix 2 for a table containing the generators having contribution to this flowgate.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

1. Reinforcements to increase the emergency rating of the Middletown Tap to Mount Pleasant 138 kV line require the rebuild of a small section of the circuit, including the installation of new poles and the re-mounting of 138 kV disconnect switches. The estimate to perform this work is \$800,000 and will take 18 months to complete.

Note: This overload was identified in a previous queue and there is a network upgrade (n5300) assigned to that project. Since the cost of the upgrade is <5M, AC1-220 doesn't get any cost allocation for the reinforcement. However, if all prior projects withdraw, the AC1 queue will need to be re-tooled and AC1-220 could get cost allocation for the reinforcement in the future.

2. Reinforcements to increase the emergency rating of the Townsend to Middletown Tap 138 kV line require the rebuild of a small section of the circuit, including the installation of new poles and the re-mounting of 138 kV disconnect switches. The estimate to perform this work is \$800,000 and will take 18 months to complete.

Note: This overload was identified in a previous queue and there is a network upgrade (n5301) assigned to that project. Since the cost of the upgrade is <5M, AC1-190 doesn't get any cost allocation for the reinforcement. However, if all prior projects withdraw, the AC1 queue will need to be re-tooled and AC1-220 could get cost allocation for the reinforcement in the future. AC1-220 isn't deliverable until the below upgrade is built.

Steady-State Voltage Requirements

None

Short Circuit

No issues identified.

Stability Analysis

No issues identified (see Attachment 2 for full report).

Light Load Analysis - 2020

Light Load Studies to be conducted during later study phases (as required by PJM Manual 14B).

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Delmarva Power and Light Costs

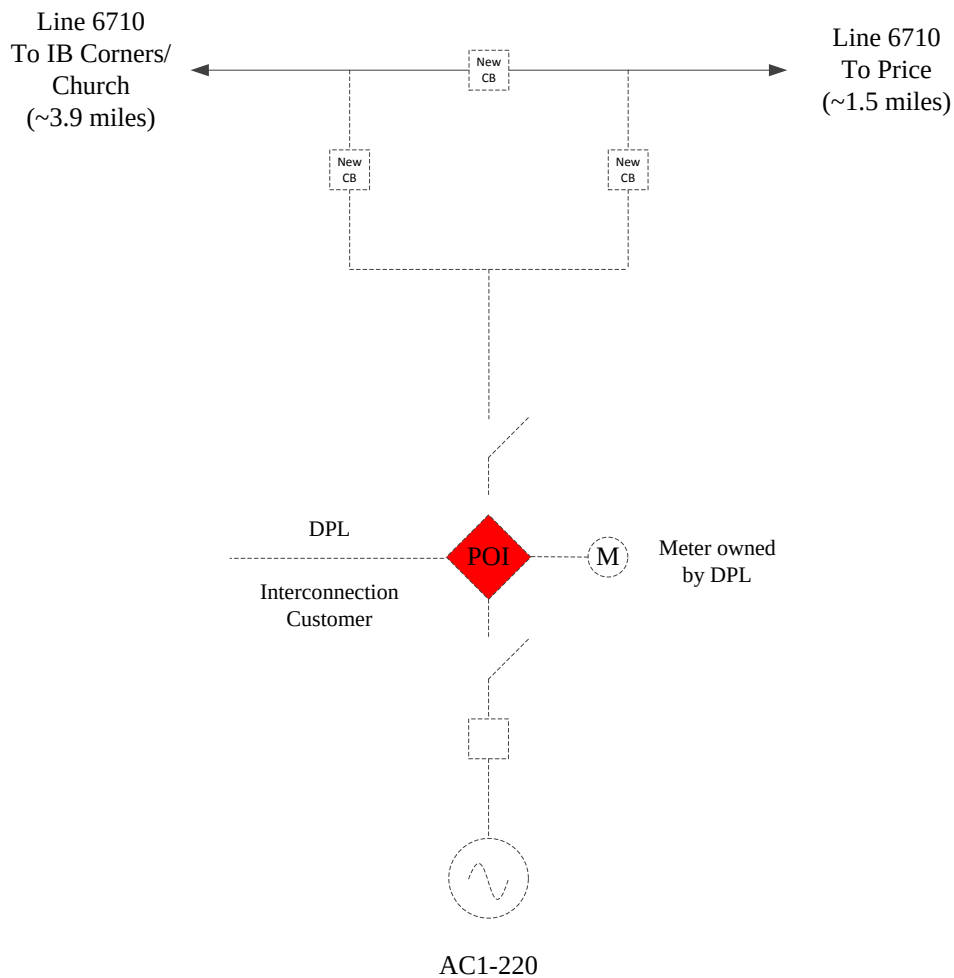
Cost estimates will further be refined as a part of the Facilities Study for this project. The Interconnection Customer will be responsible for all costs incurred by DPL in connection with the AC1-220 project. Such costs may include, but are not limited to, any transmission system assets currently in DPL's rate base that are prematurely retired due to the AC1-220 project. PJM shall work with DPL to identify these retirement costs and any additional expenses. DPL reserves the right to reassess issues presented in this document and, upon appropriate justification, submit additional costs related to the AC1-220 project.

AC1-220

Price – IB Corners (Church)

Circuit 6710-2

New 69 kV Substation



An Interconnection Customer circuit breaker will be required no more than 500 feet from the DPL substation.

 Point of Interconnection

Attachment 2

AC1-220
System Impact Study

Dynamic Simulation Analysis

Rev. 0
June 30th, 2017

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Executive Summary

Generator Interconnection Request AC1-220 is for a 26.2 MW Maximum Facility Output (MFO) solar generation plant. AC1-220 consists of 10 x TMEIC 2.7 MVA inverters with a Point of Interconnection (POI) on the Price - IB Corners 69 kV circuit, in the Delmarva Power and Light (DPL) transmission system, Queen Anne County, Maryland.

This report describes a dynamic simulation analysis of AC1-220 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2020 Summer Peak case, modified to include applicable queue projects. AC1-220 has been dispatched online at maximum power output, with approximately unity power factor at the POI.

The AC1-220 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

No relevant High Speed Reclosing (HSR) contingencies were identified.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the remaining fault contingencies tested on the 2020 Summer Peak case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AC1-220 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

1. Introduction

Generator Interconnection Request AC1-220 is for a 26.2 MW Maximum Facility Output (MFO) solar generation plant. AC1-220 consists of 10 x TMEIC 2.7 MVA inverters with a Point of Interconnection (POI) on the Price - IB Corners 69 kV circuit, in the Delmarva Power and Light (DPL) transmission system, Queen Anne County, Maryland.

This analysis is effectively a screening study to determine whether the addition of AC1-220 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AC1-220 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Request AC1-220 is for a 26.2 MW Maximum Facility Output (MFO) solar generation plant. AC1-220 consists of 10 x TMEIC 2.7 MVA inverters with a Point of Interconnection (POI) on the Price - IB Corners 69 kV circuit, in the Delmarva Power and Light (DPL) transmission system, Queen Anne County, Maryland.

Figure 1 shows the simplified one-line diagram of the AC1-220 loadflow model. Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AC1-220 loadflow model.

The dynamic model for the AC1-220 plant is based on the TMEIC L2700GR standard PSS/E user models as indicated by the developer in the System Impact Study Datasheet.

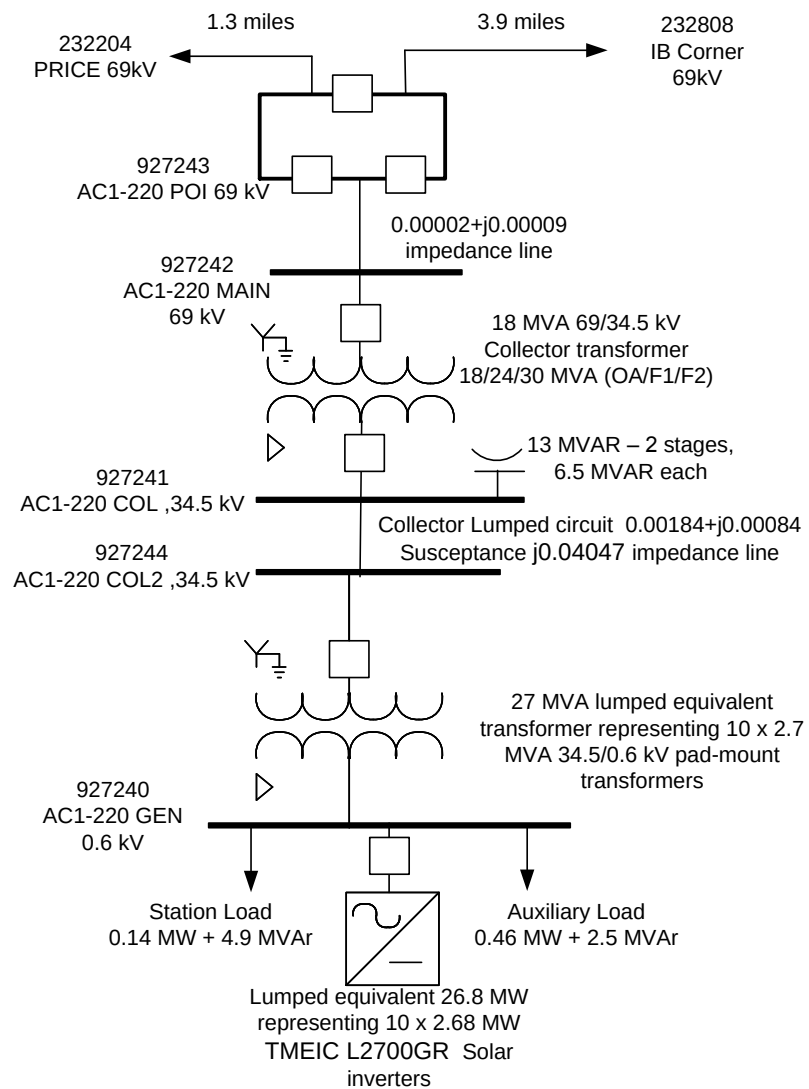


Figure 1: AC1-220 Plant Model

Table 1: AC1-220 Plant Model

	Impact Study Data	Model
Solar inverters	10 x 2.68 MW TMEIC L2700GR inverters MVA base = 2.7 MVA Vt = 0.6 kV Unsaturated sub-transient reactance = N/A	Lumped equivalent representing 10 x 2.68 MW TMEIC L2700GR inverters Pgen 26.8 MW Pmax 26.8 MW Pmin 0 MW Qgen 1.183 MVar Qmax 3.28 MVar Qmin -3.28 MVar Mbase 27 MVA Zsource¹ j 9999.0 pu @ Mbase
Solar inverter GSU transformers	10 x 34.5/0.6 kV two winding transformers Rating = 2.7 MVA Transformer base = 2.7 MVA Impedance = 0.00713 + j0.05704 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %	Lumped equivalent representing 10 x 34.5/0.6 kV two winding transformers Rating = 2.7 MVA Transformer base = 27.0 MVA Impedance = 0.00713 + j0.05704 pu @ MVA base Number of taps = 5 Tap step size = 2.5 %
Collector step-up transformer	1 x 69/34.5 kV two winding transformer Rating = 18/24/30 MVA (ONAN/ONAF/ONAF) Transformer base = 18 MVA Impedance = 0.00333 + j0.0999 pu @ MVA base Number of taps = 5 Tap step size = 2.5%	1 x 69/34.5 kV two winding transformer Rating = 18/24/30 MVA (ONAN/ONAF/ONAF) Transformer base = 18 MVA Impedance = 0.00333 + j0.0999 pu @ MVA base Number of taps = 5 Tap step size = 2.5%
Auxiliary load	0.46 MW + 2.5 MVar	0.46 MW + 2.5 MVar
Station load	0.14 MW + 4.9 MVar	0.14 MW + 4.9 MVar
Shunt Capacitor Bank	34.5kV, 13 MVAR – 2 stages, 6.5 MVAR each	34.5kV, 13 MVAR – 2 stages, 6.5 MVAR each ²

¹ ZSource = j9999 was confirmed by PJM

² Capacitor bank information presented in AC1-220 one line diagram “ Price-Roberts_Church-Hill, MD _PreSales_Rev1_2017-03-31_E1.1.pdf”

Collector Lumped Circuit Line ³	Impedance=0.00184 + 0.00084 pu Susceptance = j0.04047 pu @ 100 MVA base	Impedance=0.00184 + 0.00084 pu Susceptance = j0.04047 pu @ 100MVA base
Transmission line	Impedance=0.00002 + j0.00009 pu Susceptance = j0.0 pu	Impedance=0.00002 + j0.00009 pu Susceptance = j0.0 pu

³ Collector bus equivalent impedance is as stated in the SIS form filename:” Planning Center - Submission Admin_jsf.mht”- section “Behind the meter” under Comments.

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process⁴.

The selected load flow scenario is the RTEP 2020 Summer Peak case with the following modifications:

- a) Addition of all applicable queue projects prior to AC1-220.
- b) Addition of AC1-220 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AC1-220.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AC1-220 initial conditions are listed in Table 2, indicating maximum power output, with AC1-220 regulating to unity power factor at the generator bus.

Table 2: AC1-220 machine initial conditions

Bus	Name	Unit	PGEN (MW)	QGEN (MVAR)	ETERM (p.u.)	POI Voltage (p.u.)
927240	AC1-220 GEN 0.6000	1	26.8	-2.75	1.00	1.0145

Generation within the vicinity of AC1-220 has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AC1-220 is given in Attachment 3.

⁴ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. No relevant High Speed Reclosing (HSR) contingencies were studied.

5. Evaluation Criteria

This study is focused on AC1-220, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AC1-220 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AC1-220 is able to ride through faults (except for faults where protective action trips AC1-220).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

Due to the frequency protection was disabled due to the PSSE deficiency in calculating frequencies.

For the fault contingencies tested in this study:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AC1-220 generator was able to withstand all contingencies.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. Mitigations

No Mitigations were found to be required.

Table 3: Fault list**P0: Steady State**

Fault ID	Duration
P0.00	Steady State 20 sec run

P1: Three Phase Faults with normal clearing

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.00	3ph @ AB2-063_AB1-162 POI – Price 69kV line, normal clear loss of AB2-063 and AB1-162	9	Stable
P1.01	3ph @ Chestertown – McCleans – Lynch – Kennedyville – Massey – Church 69kV line, normal clear loss of AA1-110	9	Stable
P1.02	3ph @ Chestertown – Clough – Church 69kV line, normal clear	9	Stable
P1.03	3ph @ AB2-036 – Oil City – Steele 138kV line, normal clear	9	Stable
P1.04	3ph @ Church 69/138kV Tx #1, normal clear	9	Stable
P1.05	3ph @ Church 69/138kV Tx #2, normal clear	9	Stable
P1.06	3ph @ Church – AB2-135 POI, normal clear	9	Stable
P1.07	3ph @ Church – I.B. Corners – AC1-220 69kV line, normal clear	9	Stable
P1.08	3ph @ Centreville – Wye Mills 69kV line, normal clear	9	Stable
P1.09	3ph @ Price – AC1-220 69kV line, normal clear	9	Stable
P1.10	3ph @ Price – Centreville 69kV line, normal clear	9	Stable
P1.11	3ph @ Wye Mills – Stevensville – Grasonville 69kV line, normal clear	9	Stable
P1.12	3ph @ Wye Mills 69/138kV Tx 2, normal clear	9	Stable
P1.13	3ph @ Wye Mills – Grasonville 69kV line, normal clear	9	Stable
P1.14	3ph @ Wye Mills 69kV Cap, normal clear	9	Stable
P1.15	3ph @ Wye Mills – Longwood – Easton 69kV line, normal clear	9	Stable
P1.16	3ph @ Wye Mills 69/25kV Tx, normal clear loss of Y1-079	9	Stable

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.17	3ph @ Wye Mills 69/138kV Tx 1, normal clear	9	Stable
P1.18	3ph @ Price 69/25 kV Tx T1, normal clear loss of X3-066 and 4MW of AB1-176	9	Stable
P1.19	3ph @ Price 69/25 kV Tx T2, normal clear loss of 5MW of AB1-176	9	Stable
P1.20	3ph @ AC1-220 – I.B. Corners – Church 69kV line, normal clear	9	Stable
P1.21	3ph @ AC1-220 – Price 69kV line, normal clear	9	Stable
P1.22	3ph @ AC1-220 Main – AC1-220 POI 69kV, normal clear loss of AC1-220	9	Stable

P4: SLG Stuck Breaker (SB) Faults at Backup Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.01	SLG @ Chestertown 69/25kV Tx #1, SB @ Chestertown, delayed clear loss of Chestertown – McCleans – Lynch – Kennedyville – Massey – Church 69kV line, AA1-110	9 / 22	Stable
P4.02	SLG @ Chestertown – McCleans – Lynch – Kennedyville – Massey – Church 69kV line, normal clear loss of AA1-110, SB @ Chestertown, delayed clear loss of AB2-063	9 / 22	Stable
P4.03	SLG @ Chestertown 69/25kV Tx #2, SB @ Chestertown, delayed clear loss of Chestertown – Clough – Church 69kV line	9 / 22	Stable
P4.04	SLG @ Chestertown – Clough – Church 69kV line, SB @ Chestertown, delayed clear loss of AB2-063	9 / 22	Stable
P4.05	SLG @ Chestertown 69/25 kV Tx #1, SB @ Chestertown, delayed clear loss of Chestertown 69/25kV Tx #2	9 / 22	Stable
P4.06	SLG @ Church – AB2-036 138kV line, SB @ Church, delayed clear loss of Church – AB1-141 138kV line.	9 / 21	Stable
P4.07	SLG at Church 69 kV on Massey – Kennedyville – Lynch – McCleans – Chestertown circuit 6727 normal clear loss of AA1-110. Breaker 7210 stuck. Fault cleared with loss of Church 69 kV / 138 kV transformer AT2.	9 / 22	Stable
P4.08	SLG at Church 69 kV on Massey – Kennedyville – Lynch – McCleans – Chestertown circuit 6727, normal clear loss of AA1-110. Breaker 7290 stuck. Fault cleared with loss of Church – AB2-135 POI circuit 6704.	9 / 22	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.09	SLG at Church 69 kV on Clough – Chestertown circuit 6773. Breaker 7240 stuck. Fault cleared with loss of I.B. Corners – AC1-220 circuit 6710.	9 / 22	Stable
P4.10	SLG at Church 69 kV on Clough – Chestertown circuit 6773. Breaker 7260 stuck. Fault cleared with loss of Church 69 kV / 138 kV Transformer AT1.	9 / 22	Stable
P4.11	SLG at Church 69 kV on AB2-135 POI circuit 6704. Breaker 7290 stuck. Fault cleared with loss of Church 69 kV on Massey – Kennedeyville – Lynch – McCleans – Chestertown circuit 6727, loss AA1-110.	9 / 22	Stable
P4.12	SLG at Church 69 kV on AB2-135 POI circuit 6704. Breaker 7220 stuck. Fault cleared with loss of Church 69 kV / 25 kV Transformer T3 and Church 69 kV / 25 kV Transformer T4, Z1-081.	9 / 22	Stable
P4.13	SLG at Church 69 kV on I.B. Corners – AC1-220 circuit 6710. Breaker 7250 stuck. Fault cleared with loss of Church 69 kV / 25 kV Transformer T3 and Church 69 kV / 25 kV Transformer T4, Z1-081.	9 / 22	Stable
P4.14	SLG at Church 69 kV on I.B. Corners – AC1-220 circuit 6710. Breaker 7240 stuck. Fault cleared with loss of Clough – Chestertown circuit 6773.	9 / 22	Stable
P4.15	SLG at Church 69 kV on 69 kV / 138 kV transformer AT1. Breaker 60 stuck. Fault cleared with loss of Church 69 kV / 138 kV Transformer AT2.	9 / 22	Stable
P4.16	SLG at Church 69 kV on 69 kV / 138 kV transformer AT1. Breaker 7260 stuck. Fault cleared with loss of Clough – Chestertown circuit 6773.	9 / 22	Stable
P4.17	SLG at Church 69 kV on 69 kV / 138 kV transformer AT2. Breaker 60 stuck. Fault cleared with loss of Church 69 kV / 138 kV Transformer AT1.	9 / 22	Stable
P4.18	SLG at Church 69 kV on 69 kV / 138 kV transformer AT2. Breaker 7210 stuck. Fault cleared with loss of Church 69 kV on Massey – Kennedeyville – Lynch – McCleans – Chestertown circuit 6727. Trips AA1-110.	9 / 22	Stable
P4.19	SLG at Wye Mills – Stevensville – Grasonville 69kV line, SB @ Wye Mills, delayed clear loss of Wye Mills – Centreville 69kV line.	9 / 22	Stable
P4.20	SLG at Wye Mills – Stevensville – Grasonville 69kV line, SB @ Wye Mills, delayed clear loss of Wye Mills 69/138kV Tx 2.	9 / 22	Stable
P4.21	SLG at Wye Mills – Grasonville 69kV line, SB @ Wye Mills, delayed clear loss of Wye Mills 69/138kV Tx 2.	9 / 22	Stable

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.22	SLG at Wye Mills 69kV capbanks, SB @ Wye Mills, delayed clear loss of Wye Mills – Gransonville 69kV line.	9 / 22	Stable
P4.23	SLG at Wye Mills 69kV capbanks, SB @ Wye Mills, delayed clear loss of Wye Mills – Longwood – Easton 69kV line.	9 / 22	Stable
P4.24	3ph @ Wye Mills 69/25kV Tx, normal clear loss of Y1-079, SB @ Wye Mills, delayed clear loss of Wye Mills – Longwood – Easton 69kV line.	9 / 22	Stable
P4.25	3ph @ Wye Mills 69/25kV Tx, normal clear loss of Y1-079, SB @ Wye Mills, delayed clear loss of Wye Mills 69/138kV Tx 1.	9 / 22	Stable
P4.26	3ph @ Wye Mills 69/138kV Tx 1, SB @ Wye Mills, delayed clear loss of Wye Mills – Centreville – Price 69kV.	9 / 22	Stable
P4.27	SLG at Price – AC1-220 69kV line, SB @ Price, delayed clear loss of Price – Centreville 69kV line, loss of the X3-066, AB1-176, AB1-162, AB2-063	9 / 22	Stable
P4.28	SLG at Price – Centreville 69kV, SB @ Price, delayed clear loss of AB1-162 and AB2-063	9 / 22	Stable
P4.29	SLG at Price 69/25 kV Tx T1, normal clear loss of X3-066 and 4MW of AB1-176, SB @ Price 69kV, delayed clear loss of Price – AC1-220 69kV line	9 / 22	Stable
P4.30	SLG at Price 69/25 kV Tx T1, normal clear loss of X3-066 and 4MW of AB1-176, SB @ Price 69kV, delayed clear loss of Price 69/25 kV Tx T2, loss of 5MW of AB1-176	9 / 22	Stable
P4.31	SLG at Price 69/25 kV Tx T2, normal clear loss of 5MW of AB1-176, SB @ Price 69kV, delayed clear loss of AB2-063_AB1-162 POI – Price 69kV line, loss of AB2-063 and AB1-162	9 / 22	Stable
P4.32	SLG @ AC1-220 – Price 69kV line, SB @ AC1-220, delayed clear loss of AC1-220	9 / 22	Stable
P4.33	SLG @ AC1-220 – I.B. Corners – Church 69kV line, SB @ AC1-220, delayed clear loss of AC1-220	9 / 22	Stable
P4.34	SLG @ AC1-220 – Price 69kV line, SB @ AC1-220, delayed clear loss of AC1-220 – I.B. Corners – Church 69kV line, AC1-220	9 / 22	Stable

P5: SLG Fault with Delayed (Zone 2) Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
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Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P5.01	SLG at 80% of 138kV line from AB2-036 to Oil City – Steele, delayed clear at AB2-036.	9 / 37	Stable
P5.02	SLG at 80% of 69 kV line from Chestertown to Clough – Church circuit 6773. Delayed clearing at Chestertown.	9 / 42	Stable
P5.03	SLG at 80% of 69 kV line from Church to Massey – Kennedyville – Lynch – Chestertown circuit 6727. Delayed clearing at Church, loss of AA1-110.	9 / 42	Stable
P5.04	SLG at 80% of 69 kV line from Church to Clough – Chestertown circuit 6773. Delayed clearing at Church.	9 / 42	Stable
P5.05	SLG at 80% of 69 kV line from Church on AB2-135 POI circuit 6704. Delayed clearing at Church.	9 / 42	Stable
P5.06	SLG at 80% of 69 kV line from Church on I.B. Corners – AC1-220 circuit 6710. Delayed clearing at Church.	9 / 42	Stable
P5.07	SLG at 80% of 69kV line from Chestertown – Lynch – Kennedyville – Massey – Church, Delayed clearing at Chestertown, loss of AA1-110.	9/42	Stable
P5.08	SLG at Church 69/138kV transformer AT1, Delayed clearing	/42	Stable
P5.09	SLG at Wye Mills 69/139kV transformer AT1, delayed clearing	/42	Stable
P5.10	SLG at 80% of Wye Mills – Stevensville – Grasonville 69kV line, delayed clearing at Wye Mills	9 / 42	Stable
P5.11	SLG at 80% of Wye Mills – Grasonville 69kV line, delayed clearing at Wye Mills	9 / 42	Stable
P5.12	SLG at 80% of Wye Mills – Longwood – Easton 69kV line, delayed clearing at Wye Mills	9 / 42	Stable
P5.13	SLG at 80% of Wye Mills – Centreville 69kV line, delayed clearing at Wye Mills	9 / 42	Stable
P5.14	SLG at 80% of Centreville – Wye Mills 69kV line, delayed clearing at Centreville	9 / 42	Stable
P5.15	SLG at 80% of Price – Centreville 69kV line, delayed clearing at Price	9 / 42	Stable
P5.16	SLG at 80% of Centreville – Price 69kV line, delayed clearing at Centreville	9 / 42	Stable
P5.17	SLG at 80% of Price – AC1-220 69kV line, delayed clearing at Price	9 / 42	Stable
P5.18	SLG at Price 69/25 kV Tx T1, delayed clear, loss of X3-066 and 4MW of AB1-176	/ 42	Stable
P5.19	SLG at Price 69/25 kV Tx T2, delayed clear, loss of 5MW of AB1-176	/ 42	Stable
P5.20	SLG at 80% of AC1-220 – I.B. Corners – Church 69kV line, delayed clear at AC1-220	9/42	Stable
P5.21	SLG at 80% of AC1-220 – Price 69kV line, delayed clear at AC1-220	9/42	Stable

Attachment 2. AC1-220 PSS/E Dynamic Model

/**/*****

/**/ *** Project: AC1-220 - 26.2 MW MFO

/**/ *** POI: Tap into the Price to IB Corner 69kV Line

/**/ *** Inverter: TMEIC L2700GR inverters

/**/ *** Size: 10 x 2.68 MW Solar PV

/**/ *** PSSE Version 33

/**/*****

927240,'USRMDL', 1, 'REGCAU1', 101, 1, 1, 14, 3, 4, 1, 0.2, 10.0, 0.75,-10.0, 0.23, 2.0, 0.1, 0.0, -0.377, 0.02, 0.0, 10.0, -10.0, 0.0/

/* IBUS,'USRMDL',ID, 'REGCAU1', 101, 1, 1, 14, 3, 4,LV, Tg,Rrpwr,Brkpt, Zrox,Lvpl1,Volm,Lvpt,Lvp0, Iolim,Tfltr, Khv,Iqrmx,Iqrmin,Accl

927240,'USRMDL', 1, 'REECBU1', 102, 0, 5, 25, 6, 4, 0, 0, 0, 0, 0, 0.0, 2.0, 0.0, -0.1, 0.1, 0.0, 0.377, -0.377, 0.0, 0.05, 0.377, -0.377, 1.1, 0.9, 0.0, 0.0, 0.0, 0.0, 0.02, 2.0, -2.0, 0.993, 0.0, 1.09, 0.02/

/* IBUS,'USRMDL',ID, 'REECBU1', 102, 0, 5, 25, 6, 4,Vb,PF,VF,QF,PQ,Vdip, Vup, Trv, dbd1,dbd2, Kqv, iqh1, iql1,Vrf0, Tp, QMax, QMin,Vmax,Vmin, Kqp, Kqi, Kvp, Kvi, Tiq,dPmx,dPmin, Pmax,PMin, imax, Tprd

927240,'USRMDL', 1, 'REPCAU1', 107, 0, 7, 27, 7, 9, 927243, 0, 0, 0, 0, 1, 0, 0.02, 18, 5, 0, 0.15, -1, 0, 0, 0, 999, -999,-0.02, 0.02, 0.377, -0.377, 10, 1, 0.02, -99.0, 99.0, 999, -999, 0.993, 0, 20, 20, 20/

/* IBUS,'USRMDL',ID, 'REPCAU1', 107, 0, 7, 27, 7, 9, Vb,FB,Tb,ID,VF,RF,FF,Tfltr, Kp,Ki,Tf, Tfv,Vfz,Rc,Xc,Kc,emax, emin, d1, d2, Qmax, Qmin, Kp,Kg, Tp, fdbd1,Fdbd2,fmax, fmin, Pmax,Pm, Tg,Ddn,Dup

9272401, 'VTGTPAT', 927243, 927240, 1, -1, 1.200, 0, 0.0/

9272402, 'VTGTPAT', 927243, 927240, 1, -1, 1.175, 0.2, 0.0/

9272403, 'VTGTPAT', 927243, 927240, 1, -1, 1.15, 0.5, 0.0/

9272404, 'VTGTPAT', 927243, 927240, 1, -1, 1.10, 1.0, 0.0/

9272405, 'VTGTPAT', 927243, 927240, 1, 0.45, 5, 0.20, 0.0/

9272406, 'VTGTPAT', 927243, 927240, 1, 0.65, 5, 0.80, 0.0/

9272407, 'VTGTPAT', 927243, 927240, 1, 0.75, 5, 2, 0.0/

9272408, 'VTGTPAT', 927243, 927240, 1, 0.90, 5, 3, 0.0/

9272409, 'FRQTPAT', 927243, 927240, 1, -100, 61.8, 0, 0.0/

9272410, 'FRQTPAT', 927243, 927240, 1, -100, 60.5, 600.66, 0.0/

9272412, 'FRQTPAT', 927243, 927240, 1, 57.8, 100, 0, 0.0/

9272413, 'FRQTPAT', 927243, 927240, 1, 59.5, 100, 1792.049, 0.0/

Attachment 3. AC1-220 PSS/E Case Dispatch

Bus Number	Bus Name	Id	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)
232227	EASTN_69 69.000	1	1	0	0	0	-30	0	-30
232813	VAUGHN 69.000	1	1	3	3	0	0	0	0
232851	DUP-SFR1 12.470	1	1	10	10	0	0	0	0
232902	EASTMUNI 69.000	1	1	69	69	0	34.6	34.6	0
232907	VN8 18.000	8	1	153	153	0	-15	28	-15
232919	VN10 13.200	1	1	14.3	14.3	0	0	14.3	0
297076	V2-028 C 69.000	1	1	2.28	2.28	0	0	0	0
297077	V2-028 E 69.000	1	1	3.72	3.72	0	0	0	0
910571	X3-008 C 69.000	1	1	7.6	7.6	0	0	0	0
910572	X3-008 E 69.000	1	1	12.4	12.4	0	0	0	0
910821	X3-066 C 25.000	1	1	2.28	2.28	0	0	0	0
910822	X3-066 E 25.000	1	1	3.72	3.72	0	0	0	0
913361	Y1-079 C 24.900	1	1	3.8	3.8	0	0	0	0
913362	Y1-079 E 24.900	1	1	6.2	6.2	0	0	0	0
916281	Z1-081 C 24.900	1	1	2.28	2.28	0	0	0	0
916282	Z1-081 E 24.900	1	1	3.72	3.72	0	0	0	0
923923	AB2-032 GEN 0.6000	1	1	20	20	0	2.226	6.58	-6.58
923953	AB2-036 GEN 0.3850	1	1	102.4	102.4	0	-9	33.66	-33.7
924191	AB2-063 GEN 0.4180	1	1	20	20	0	-6.57	6.573	-6.57
924804	AB2-133 GEN 0.6000	1	1	57.04	57.04	0	5.591	18.8	-18.8
924824	AB2-135 GEN 0.6000	1	1	65.4	65.4	0	9.208	21.5	-21.5
924973	AB2-153 GEN 0.6000	1	1	20	20	0	2.226	6.58	-6.58
925253	AB2-179 GEN 0.3850	1	1	50	50	0	6.053	16.5	-16.5
925271	AB2-185 C OP24.900	1	1	14	14	0	0	0	0
925272	AB2-185 E OP24.900	1	1	6	6	0	0	0	0
926721	AC1-154 C 69.000	1	1	1.2	1.2	0	0.396	0.396	-0.4
926722	AC1-154 E 69.000	1	1	2	2	0	0.396	0.396	-0.4
927240	AC1-220 GEN 0.6000	1	1	26.8	26.8	0	-3.28	3.28	-3.28
927272	AC1-224 E 69.000	1	0	2	2	0	0	0	0
930922	AB1-141 GEN 0.5500	1	1	20	20	0	2.226	6.58	-6.58
930932	AB1-142 GEN 0.5500	1	1	20	20	0	2.226	6.58	-6.58
931111	AB1-162 GEN 0.4180	1	1	16.7	16.7	0	-5.48	5.479	-5.48
931261	AB1-176 GEN 0.4800	1	1	5	5	0	0	0	0
931262	AB1-176 GEN 0.4800	1	1	4	4	0	0	0	0

Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact.

It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

Appendix 1

(DP&L - DP&L) The MIDLTNTP-MT PLSNT 138 kV line (from bus 232106 to bus 232104 ckt 1) loads from 100.85% to 103.49% (AC power flow) of its emergency rating (348 MVA) for the tower line contingency outage of 'DBL_4NC'. This project contributes approximately 10.12 MW to the thermal violation.

CONTINGENCY 'DBL_4NC'/* RED LION-CEDAR CREEK 230;RED LION-CARTANZA 230
OPEN LINE FROM BUS 231004 TO BUS 232002 CKT 1
OPEN LINE FROM BUS 231004 TO BUS 232003 CKT 1

<i>ENDBus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
232900	DEMECSMY	2.11
232851	DUP-SFR1	0.4
232902	EASTMUNI	3.34
232923	MR1	3.37
232924	MR2	3.37
232910	NRG_G1	2.39
232911	NRG_G2	2.39
292089	T-011	0.16
297076	V2-028 C	0.46
297077	V2-028 E	0.75
904212	V4-022E	0.61
232813	VAUGHN	0.15
232919	VN10	0.56
901004	W1-003 E	0.89
901014	W1-004 E	0.89
901024	W1-005 E	0.89
901034	W1-006 E	0.89
901411	W1-062	2.24
907052	X1-032 E	0.79
907324	X1-096 E	18.29
910571	X3-008 C	0.32

910572	X3-008 E	2.68
910591	X3-015 C	1.54
910592	X3-015 E	2.52
910821	X3-066 C	0.17
910822	X3-066 E	1.41
913361	Y1-079 C	0.23
913362	Y1-079 E	1.96
913411	Y1-080 C	0.05
913412	Y1-080 E	0.43
915541	Y3-058 C	0.22
915542	Y3-058 E	1.86
920582	Z1-076 C	1.05
920583	Z1-076 E	1.71
920592	Z1-077 C	0.75
920593	Z1-077 E	1.22
917082	Z2-012 E	2.45
920763	Z2-076 E	0.4
920773	Z2-077 E	0.4
920812	Z2-097 C	0.3
920813	Z2-097 E	0.65
921123	AA1-059 E	0.33
921142	AA1-061 C	2.88
921143	AA1-061 E	1.42
921442	AA1-110 C	0.34
921443	AA1-110 E	0.89
921592	AA1-140 C	1.52
921593	AA1-140 E	2.47
921602	AA1-141 C	1.13
921603	AA1-141 E	1.84
921872	AA2-069	104.95
922213	AA2-129 E	3.94
922222	AA2-130	0.39
922752	AB1-056 C OP	12.81
922753	AB1-056 E OP	36.48
922762	AB1-057 C	13.01
922763	AB1-057 E	37.08
923282	AB1-137 C	2.79
923283	AB1-137 E	1.2
923322	AB1-141 C OP	5.3
923323	AB1-141 E OP	2.47
923332	AB1-142 C OP	5.3
923333	AB1-142 E OP	2.47
923452	AB1-162 C OP	2.4
923453	AB1-162 E OP	3.92
923602	AB1-176 C	1.29

923603	AB1-176 E	2.12
923902	AB2-030 E	0.79
923921	AB2-032 C	5.34
923922	AB2-032 E	2.51
923931	AB2-033 C	1.41
923932	AB2-033 E	0.56
923951	AB2-036 C	11.45
923952	AB2-036 E	18.74
923961	AB2-037 C	12.75
923962	AB2-037 E	20.83
924191	AB2-063 C	2.87
924192	AB2-063 E	4.69
924361	AB2-084 C	0.75
924362	AB2-084 E	1.22
924681	AB2-120 C OP	7.5
924682	AB2-120 E OP	12.23
924781	AB2-130 C OP	6.58
924782	AB2-130 E OP	10.74
924801	AB2-133 C OP	10.92
924802	AB2-133 E OP	13.86
924821	AB2-135 C	12.85
924822	AB2-135 E	14.65
924831	AB2-136 C	1.
924832	AB2-136 E	5.52
924971	AB2-153 C	2.98
924972	AB2-153 E	4.87
925091	AB2-166 C	0.4
925092	AB2-166 E	0.7
925101	AB2-167 C	1.05
925102	AB2-167 E	1.72
925151	AB2-172 C OP	4.11
925152	AB2-172 E OP	6.71
925231	AB2-177 C	0.49
925232	AB2-177 E	0.81
925251	AB2-179 C OP	26.3
925252	AB2-179 E OP	8.67
925261	AB2-180 C	2.8
925262	AB2-180 E	1.2
925271	AB2-185 C OP	1.77
925272	AB2-185 E OP	0.76
926081	AC1-009 C	1.5
926082	AC1-009 E	2.45
926401	AC1-041 C	0.38
926402	AC1-041 E	0.61
926481	AC1-049 C	0.3

926482	AC1-049 E	0.49
926491	AC1-050 C	0.38
926492	AC1-050 E	0.61
926511	AC1-052 C	1.26
926512	AC1-052 E	0.51
926931	AC1-094 C	6.18
926932	AC1-094 E	10.15
926941	AC1-095 C	0.84
926942	AC1-095 E	1.34
927641	AC1-154 C	0.49
927642	AC1-154 E	0.8
927871	AC1-177	0.79
928001	AC1-190 C	7.53
928002	AC1-190 E	3.23
928241	AC1-213 C	0.64
928242	AC1-213 E	0.42
928311	AC1-220 C	5.95
928312	AC1-220 E	4.17
928352	AC1-224 E	0.32
928391	AC1-228 C	0.24
928392	AC1-228 E	0.4
928401	AC1-229 C	0.76
928402	AC1-229 E	1.21

Appendix 2

(DP&L - DP&L) The TOWNSEND-MIDLTNTP 138 kV line (from bus 232107 to bus 232106 ckt 1) loads from 111.18% to 113.81% (AC power flow) of its emergency rating (348 MVA) for the tower line contingency outage of 'DBL_4NC'. This project contributes approximately 10.12 MW to the thermal violation.

CONTINGENCY 'DBL_4NC'/* RED LION-CEDAR CREEK 230;RED LION-CARTANZA 230
 OPEN LINE FROM BUS 231004 TO BUS 232002 CKT 1
 OPEN LINE FROM BUS 231004 TO BUS 232003 CKT 1
 END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
232900	DEMECSMY	2.11
232851	DUP-SFR1	0.4
232902	EASTMUNI	3.34
232923	MR1	3.37
232924	MR2	3.37
232910	NRG_G1	2.39
232911	NRG_G2	2.39
292089	T-011	0.16
297076	V2-028 C	0.46
297077	V2-028 E	0.75
904212	V4-022E	0.61
232813	VAUGHN	0.15
232919	VN10	0.56
901004	W1-003 E	0.89
901014	W1-004 E	0.89
901024	W1-005 E	0.89
901034	W1-006 E	0.89
901411	W1-062	2.24
907052	X1-032 E	0.79
907324	X1-096 E	18.29
910571	X3-008 C	0.32
910572	X3-008 E	2.68
910591	X3-015 C	1.54
910592	X3-015 E	2.52
910821	X3-066 C	0.17
910822	X3-066 E	1.41
913361	Y1-079 C	0.23
913362	Y1-079 E	1.96
913411	Y1-080 C	0.05
913412	Y1-080 E	0.43
915541	Y3-058 C	0.22

915542	Y3-058 E	1.86
920582	Z1-076 C	1.05
920583	Z1-076 E	1.71
920592	Z1-077 C	0.75
920593	Z1-077 E	1.22
917082	Z2-012 E	2.45
920763	Z2-076 E	0.4
920773	Z2-077 E	0.4
920812	Z2-097 C	0.3
920813	Z2-097 E	0.65
921123	AA1-059 E	0.33
921142	AA1-061 C	2.88
921143	AA1-061 E	1.42
921442	AA1-110 C	0.34
921443	AA1-110 E	0.89
921592	AA1-140 C	1.52
921593	AA1-140 E	2.47
921602	AA1-141 C	1.13
921603	AA1-141 E	1.84
921872	AA2-069	104.95
922213	AA2-129 E	3.94
922222	AA2-130	0.39
922752	AB1-056 C OP	12.81
922753	AB1-056 E OP	36.48
922762	AB1-057 C	13.01
922763	AB1-057 E	37.08
923282	AB1-137 C	2.79
923283	AB1-137 E	1.2
923322	AB1-141 C OP	5.3
923323	AB1-141 E OP	2.47
923332	AB1-142 C OP	5.3
923333	AB1-142 E OP	2.47
923452	AB1-162 C OP	2.4
923453	AB1-162 E OP	3.92
923602	AB1-176 C	1.29
923603	AB1-176 E	2.12
923902	AB2-030 E	0.79
923921	AB2-032 C	5.34
923922	AB2-032 E	2.51
923931	AB2-033 C	1.41
923932	AB2-033 E	0.56
923951	AB2-036 C	11.45
923952	AB2-036 E	18.74
923961	AB2-037 C	12.75
923962	AB2-037 E	20.83

924191	AB2-063 C	2.87
924192	AB2-063 E	4.69
924361	AB2-084 C	0.75
924362	AB2-084 E	1.22
924681	AB2-120 C OP	7.5
924682	AB2-120 E OP	12.23
924781	AB2-130 C OP	6.58
924782	AB2-130 E OP	10.74
924801	AB2-133 C OP	10.92
924802	AB2-133 E OP	13.86
924821	AB2-135 C	12.85
924822	AB2-135 E	14.65
924831	AB2-136 C	1.
924832	AB2-136 E	5.52
924971	AB2-153 C	2.98
924972	AB2-153 E	4.87
925091	AB2-166 C	0.4
925092	AB2-166 E	0.7
925101	AB2-167 C	1.05
925102	AB2-167 E	1.72
925151	AB2-172 C OP	4.11
925152	AB2-172 E OP	6.71
925231	AB2-177 C	0.49
925232	AB2-177 E	0.81
925251	AB2-179 C OP	26.3
925252	AB2-179 E OP	8.67
925261	AB2-180 C	2.8
925262	AB2-180 E	1.2
925271	AB2-185 C OP	1.77
925272	AB2-185 E OP	0.76
926081	AC1-009 C	1.5
926082	AC1-009 E	2.45
926401	AC1-041 C	0.38
926402	AC1-041 E	0.61
926481	AC1-049 C	0.3
926482	AC1-049 E	0.49
926491	AC1-050 C	0.38
926492	AC1-050 E	0.61
926511	AC1-052 C	1.26
926512	AC1-052 E	0.51
926931	AC1-094 C	6.18
926932	AC1-094 E	10.15
926941	AC1-095 C	0.84
926942	AC1-095 E	1.34
927641	AC1-154 C	0.49

927642	AC1-154 E	0.8
927871	AC1-177	0.79
928001	AC1-190 C	7.53
928002	AC1-190 E	3.23
928241	AC1-213 C	0.64
928242	AC1-213 E	0.42
928311	AC1-220 C	5.95
928312	AC1-220 E	4.17
928352	AC1-224 E	0.32
928391	AC1-228 C	0.24
928392	AC1-228 E	0.4
928401	AC1-229 C	0.76
928402	AC1-229 E	1.21